

LEE KAH HIN

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SUMMARY

Cross-Platform Systems & Interactive Engineer with nearly **3 years of experience in full-stack** development , core gameplay frameworks, and responsive cloud-integrated system. Skilled in C#, Flutter, React, and Node.js, with expertise in OOP, state-driven design patterns, and decoupled micro-frontend environments. Proven record of optimizing core system algorithms, executing seamless real-time data sync, and delivering robust solutions under strict deadlines. **Seeking a Full-Stack or Game Developer role** to build innovative, scalable and customer-focused applications.

EDUCATION

Masha University Bachelor of Information Technology(HONS)	Oct 2020 – Dec 2022
Linton University College - Foundation in Science (Information Technology) - Bachelor of Software Engineering(HONS)	Jan 2018 – Jan 2019 Apr 2019-Mar 2020

WORK

EXPERIENCE

Oct 2023 – Present

Star Splash Game
Studio SDN BHD
Cross-Platform Systems
& Interactive Engineer

- Spearheaded full-stack creative engineering and interactive systems development, operating as the primary standalone engineer following management restructuring to independently drive project lifecycles from concept to deployment.
- Designed and optimized responsive UI/UX gesture detection logic and custom shader modifications in Unity for Project ALF, establishing strict client alignment through agile technical requirement translation.
- Engineered core systems for Cardcaptorh (a collectible card game), including dynamic collection displays, tailored shaders for cards, and seamless Steam API Integration for real-time gacha-to-Steam-inventory synchronization and strict authentication runtime.
- Architected complex game loop mechanics for an Auto-Chess R&D prototype, developing a data-driven faction system (e.g., modular race-based attribute modifiers) and autonomous AI combat targeting state logic.
- Embedded with a collaboration team to engineer a real-time multiplayer chat environment for a web-based title, managing robust API synchronization pipelines with remote corporate servers.
- Delivered a production-grade Mobile Travel Planner App using Flutter; orchestrated the entire deployment pipeline by performing native iOS compilation via Xcode, archiving builds, and managing TestFlight release cycles for iterative client feedback loops.
- Developed an interactive Unity system integrated with the ElevenLabs REST API for advanced voice synthesis, executing full Xcode archiving and TestFlight provisioning to ensure seamless client-side validation.
- Built the AI Storyboard Generator platform powered by Google Gemini LLM orchestration, implementing secure Google OAuth verification and Firebase database architectures for persistent data state management.
- Developed the ETI Enterprise Portal using React and Node.js; built standalone Express endpoint services to offload backend workflows, while establishing the company's production deployment pipeline via FTP/SFTP, FileZilla, and PM2 process

management.

- Integrated Google's Gemma SLM (Small Language Model) for on-device edge computing; pioneered balance-optimized prompt engineering models and resolved hard-to-trace MediaPipe ARM64 hardware multi-threading bugs by forcing main-thread inference constraints.

KidoCode

Jan 2023 – July 2023

Game Development Teacher

- Led and delivered structured programming and game development education for diverse age groups (ages 5-18), translating complex coding concepts into highly intuitive, logic-driven mental models.
- Served as a versatile full-stack troubleshooter, debugging multi-language codebases (Scratch, Python, JavaScript, and game engines) to resolve technical and curriculum system issues in real-time.
- Cultivated exceptional technical communication and cross-functional agility by managing student problem-solving pipelines and clarifying algorithmic logic on short notice.

KidoCode

June 2022 – Dec 2022

Game Development (Internship)

- Contributed to the development and testing of interactive mini-game prototypes and foundational software scripts to act as templates for educational learning environments.
- Modeled basic algorithmic workflows and step-by-step logic scenarios to ensure data integrity and structural reliability in student projects.

PROJECTS

ETI Enterprise Portal

Methodologies: Decoupled Full-Stack Architecture, Production Process Management

Technologies: React, Node.js, Express, HTML5/CSS3, Vite, FTP/SFTP, PM2, FileZilla

- Architected and delivered a multi-module enterprise website utilizing React and modern CSS, achieving fluid interfaces and highly modularized cross-platform code reuse.
- Designed and deployed standalone Node.js/Express endpoint services to handle localized data routing, successfully abstracting workflows away from core backend components.
- Established the company's secure production deployment pipeline, managing static asset distribution via FTP/SFTP protocols and executing continuous process lifecycle management using PM2 and port monitoring on remote web servers.

SoulWave AI Voice Sandbox

Methodologies: Interactive Cloud Orchestration, Asynchronous Data Streaming

Technologies: Unity Engine, C#, RESTful APIs, ElevenLabs API, Xcode, TestFlight

- Developed an immersive generative AI voice sandbox within Unity, establishing secure handshakes and synchronous query structures with remote ElevenLabs REST APIs for low-latency text-to-speech output.
- Implemented optimized asynchronous data pipelines to ingest, analyze, and stream cloud-generated audio arrays, linking computational data metrics directly to real-time interactive UI components.
- Streamlined the native deployment lifecycle by conducting comprehensive iOS builds via Xcode, archiving deliverables, and managing TestFlight internal testing tracks to rapidly incorporate iterative client specifications.

Trivium-System (On-Device Gemma SLM Optimization Hub)

Methodologies: Edge Computing Inference, Algorithmic Stabilization, Hardware Debugging

Technologies: Google Gemma 4 (SLM), Lite-LLM Plugin, MediaPipe Framework, ARM64 Architecture

- Orchestrated the client-side integration of Google's Gemma Small Language Model (SLM), enabling completely autonomous, local-device AI model caching and offline inference processing.
- Stabilized AI response formats and neutralized erratic agent behavior through batch-prompt engineering methodologies and strict early-stop character sequence detection to prevent buffer overflows.
- Overcame a critical, undocumented MediaPipe multi-threading crash inherent to ARM64 physical mobile hardware by designing thread-confinement overrides that forced AI inference execution directly onto the main application thread.

Cardcaptorh: Steam-Integrated Collectible Card Game

Methodologies: Secure Platform Integration, Algorithmic Gacha Pipelines, Client-Server Sync

Technologies: Unity Engine, C#, Steamworks API, Custom Shader Lab, JSON Systems

- Engineered the dynamic collection and inventory rendering systems for a production-grade collectible card game, securing high-performance rendering under dense UI scaling.
- Designed a secure, automated synchronization pipeline connecting local gacha draw events directly to the Steamworks API, enabling real-time Steam Inventory updates and drop-rate calculations.
- Implemented strict network authentication logic requiring validated Steam client runtimes to initialize game boot sequences, effectively preventing unauthorized client access.
- Authored custom rendering shaders within Unity Shader Lab to elevate the visual fidelity of interactive card assets, striking an optimal balance between visual output and execution performance.

ADDITIONAL PROJECTS

- **Project ALF (Unity/C#):** An R&D matrix framework managing O(1) spatial state calculations and gesture recognition.
- **Travel Planner Mobile App (Flutter/Dart):** A cross-platform mobile application compiled natively via Xcode and deployed via TestFlight.
- **Personal TCG Engine (Unity/C#):** A personal sandbox exploring dynamic deck-building algorithms and local state save-systems.

SKILLS AND LANGUAGES

Technical Skills

- **Languages & Frameworks:** C#, JavaScript, TypeScript, Flutter (Dart), React, Node.js, Express, HTML5/CSS3
- **Databases & APIs:** Firebase (Realtime DB), Google Workspace API (Sheets-as-CMS), Google Gemini API, ElevenLabs API, Steamworks API
- **Tools & Platforms:** Git, Filezilla, Unity Engine, Vite, Xcode, SourceTree, Visual Studio, PM2, .env Configuration
- **Design & UX/UI:** Figma, Draw.io
- **Methodologies:** OOP, MVC, Agile Development, Data Pipeline Automation, AI-Augmented Engineering

Soft Skills

- Problem-Solving, Communication, Critical Thinking, Multitasking & Time Management, Adaptability, Team Collaboration, Fast Learner

Languages

- Chinese (Fluent), Malay (Basic), English (Fluent), Cantonese